

Foot-Mounted IMU Estimation of Minimum Toe Clearance with Low-Cost Wearable Signal Verification

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Abstract— Minimum toe clearance (MTC) is an important gait parameter for trip-risk assessment, but conventional motion capture (MoCap) measurement is laboratory-dependent and unsuitable for routine monitoring. This study evaluates a foot-mounted inertial measurement unit (IMU) workflow for estimating MTC and investigates whether a low-cost custom wearable prototype can capture plausible gait-related inertial signals. The algorithmic pipeline was validated using the SensorPositionComparison2019 (SPC2019) dataset, which provides synchronized foot-mounted IMU and MoCap recordings. A strict-nested EventNet gait-event detector was used to identify toe-off and initial-contact events, followed by adaptive stride-local vertical integration, virtual toe kinematics, MoCap-referenced toe-offset calibration, and stride-level IMU-MoCap comparison. The final reported validation used the 12 completed participant cohort; two indexed participants were excluded from final MTC metrics because held-out ML event recovery produced no valid stride matches for two-stage MoCap-referenced toe-offset calibration under the fixed matching criteria. Across 7142 matched strides, the IMU-based estimator achieved an overall MTC bias of -3.000 mm, MAE of 8.638 mm, RMSE of 11.401 mm, and Pearson correlation of $r = 0.674$. Normal-speed walking produced the most stable agreement, while fast trials and long mixed-speed walking showed greater sensitivity to event timing and stride matching. A separate BMI160/XIAO prototype was also developed and verified at the signal level using gyroscope-norm waveform comparison against SPC2019 right-instep references. The prototype captured repeatable gait-cycle peaks and speed-dependent rhythm changes, supporting hardware signal plausibility but not direct MTC accuracy. Overall, the work supports a staged validation pathway in which the offline IMU algorithm is technically promising, while the low-cost prototype requires synchronized BMI160-MoCap testing before direct MTC validation can be claimed.

Index Terms—Gait analysis, inertial measurement units, minimum toe clearance, motion capture, wearable sensors.

I. INTRODUCTION

Minimum toe clearance (MTC) is the minimum vertical distance between the toe and walking surface during the swing phase of gait. It is often discussed under minimum foot clearance (MFC), which is associated with trip risk because the swing foot passes close to the ground during mid-swing [1]. Accurate MTC estimation is therefore relevant for gait safety, fall-risk assessment, and walking-performance analysis. Although optical motion capture (MoCap) provides accurate marker-based clearance measurements, its high cost, laboratory dependence, and limited practicality make it unsuitable for routine or real-world

monitoring [1], [2].

To address these limitations, wearable inertial measurement units (IMUs) offer a portable and lower-cost alternative to MoCap. By measuring acceleration and angular velocity, IMUs can support gait-event detection, foot-trajectory estimation, and lower-limb movement analysis [2], [3]. Previous studies have shown that foot-mounted IMUs can estimate heel, toe, and foot clearance [2], [3], while recent work has improved wearable clearance estimation using single foot-mounted IMUs combined with shoe-geometry models [4].

Despite this potential, IMU-based MTC estimation remains challenging because IMUs do not directly measure displacement. Foot or toe trajectory must be reconstructed by transforming acceleration into a reference frame and integrating it twice, making the estimate sensitive to sensor bias, orientation error, gait-event timing error, and drift [2]. These issues are particularly important because MTC occurs during swing, where errors in toe-off, final contact, or initial contact detection can alter the integration window and affect the estimated clearance. Deep-learning gait-event detection has shown good agreement with optical motion-capture references, supporting its use in IMU-based gait pipelines [5].

These challenges show that wearable clearance estimation requires not only an appropriate IMU processing method, but also careful validation against reliable reference data. Existing studies demonstrate the feasibility of wearable clearance estimation while highlighting the importance of sensor placement, calibration, drift correction, gait-event detection, and reference-based validation [2]–[4]. This project therefore uses the SensorPositionComparison2019 dataset, hereafter referred to as SPC2019 dataset because it provides synchronized foot-mounted IMU and MoCap recordings across multiple walking tests and sensor positions, allowing direct comparison between IMU-estimated and MoCap-derived MTC [6], [7].

Beyond algorithm validation, a further practical issue is whether a low-cost custom wearable IMU device can capture usable gait-related inertial signals. Many studies validate clearance algorithms using research-grade sensors or controlled laboratory setups, while fewer examine the early-stage feasibility of custom low-cost devices for this purpose. Therefore, this project separates algorithm validation from hardware verification. The MTC algorithm is validated using SPC2019 MoCap reference data, while the custom BMI160/XIAO-based prototype is assessed at the raw-signal

level by comparing its waveform morphology with reference foot-mounted IMU recordings [8], [9]. Pearson correlation, normalized root mean square error (nRMSE), cross-correlation, and dynamic time warping (DTW) are used because they assess waveform similarity under amplitude and timing differences [10].

The objective of this work is therefore to evaluate whether a foot-mounted IMU processing pipeline can estimate MTC with acceptable agreement against MoCap reference data, and whether a low-cost custom IMU prototype can capture recognizable gait-related inertial patterns. The contributions are:

- 1) Implementation of a foot-mounted IMU-based MTC pipeline using SPC2019.
- 2) Comparison of estimated MTC against synchronized MoCap data.
- 3) Development of a low-cost BMI160/XIAO wearable IMU prototype.
- 4) Preliminary signal-level verification of the prototype against reference IMU waveforms.

II. HARDWARE SYSTEM

A. Circuit Design and Sensor Integration

The custom BMI160 device was designed under low-cost, wearable-size, untethered-operation, and shoe-mounting constraints. An external BMI160 IMU was connected to a Seeed Studio XIAO nRF52840 Sense microcontroller, with the BMI160 measuring triaxial acceleration and angular velocity while the XIAO handled I2C communication, BLE transfer, and battery-supported operation [8], [9]. The circuit used SDA, SCL, 3.3 V, and ground, with a common ground maintained across the IMU, microcontroller, battery, and slide switch to support stable readings. A lithium-polymer battery and slide switch enabled untethered walking and safe isolation during mounting, handling, and storage.

B. CAD Enclosure

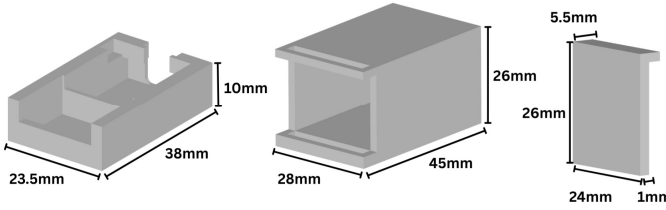


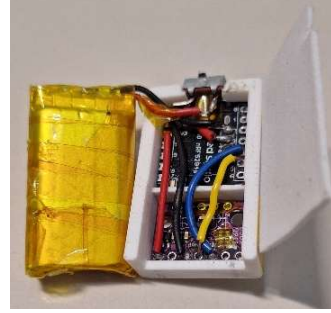
Fig. 1. CAD components of the custom BMI160/XIAO wearable enclosure, showing the bottom layer (left), outer shell (middle), and shell door (right).

A custom two-layer CAD enclosure with dimensions of $45 \times 26 \times 28$ mm was designed as a compact wearable casing. The top layer housed the lithium-polymer battery, while the bottom layer contained the BMI160 IMU, XIAO nRF52840 Sense microcontroller, slide switch, and wiring.

Mechanical stability was the main design consideration because foot-mounted IMU outputs are affected by sensor placement and fixation during walking [7]. The component slots were dimensioned to closely fit the BMI160 and

microcontroller, reducing internal movement while preserving wiring clearance, USB access, battery placement, and ease of assembly.

C. Fabrication and Assembly



After CAD design, the battery, slide switch, XIAO nRF52840 Sense microcontroller, and BMI160 IMU were soldered together to reduce loose connections during walking trials. The lithium-polymer battery was insulated with electrical tape before being placed in the top layer of the casing.



EVA foam was added inside the battery compartment and between the battery and lower electronics

layer to improve fit, reduce battery movement, and separate the battery from the microcontroller, IMU, and wiring. A 2.5 mm drill bit was also used to drill through the casing so the BMI160 could be fastened tightly, reducing relative movement between the sensor and enclosure during walking [7].

Fig. 2. Fabricated BMI160/XIAO prototype and internal assembly details.

D. Shoe Mounting



Fig. 3. Shoe-mounted wearable IMU enclosure positioned at the instep/tongue region and secured using interlaced shoelaces.

The enclosure was mounted near the shoe instep/tongue region using Velcro strap tape and shoelaces. Velcro was attached to the shoe tongue and casing base to reduce movement, while the shoelaces provided additional fastening. This position was selected because it was the closest practical match to the SPC2019 instep sensor location used for comparison [6], [7].

This mounting position balanced sensor fixation, walking comfort, accessibility, and reduced swing-phase obstruction. The same attachment method was used across slow, normal,

and fast trials to reduce variation from placement, tightness, or casing slip [7].

E. BLE Data Transfer

Bluetooth Low Energy (BLE) was used to transfer IMU data to a host computer without a physical tether, consistent with the wireless capability of the XIAO nRF52840 platform [9]. Each sample was transmitted as a structured row containing timestamp, triaxial acceleration, and triaxial angular velocity, then saved as trial-specific CSV files.

To improve transfer reliability, a simple three-way handshake was used. The host initiated recording, the device transmitted CSV data in chunks after recording stopped, and the host returned ACK/NACK responses to confirm receipt or request retransmission before the next chunk.

III. DATASET

This project used two data sources: the online SPC2019 dataset and custom recordings from the low-cost BMI160 device. SPC2019 was used because it was recorded using commercial Nilspod IMU sensors with synchronized MoCap data, making it suitable for comparison with the custom BMI160 prototype [6], [7]. The full dataset contains 14 participants [6].

A. SPC2019 Dataset

SPC2019 contains walking trials recorded with IMU sensors attached to both feet, together with synchronized MoCap data [6], [7]. Each participant has seven walking trials: slow_10, normal_10, fast_10, slow_20, normal_20, fast_20, and long [6]. The slow, normal, and fast labels refer to the participant's preferred walking speed for each condition, while 10 and 20 refer to the walking distance before turning. The long trial combines slow, normal, and fast walking within one continuous recording.

B. Selected IMU and MoCap Signals

For MTC validation, SPC2019 triaxial acceleration and angular velocity from the evaluated instep-mounted foot sensors were used [6], [7]. The corresponding MoCap foot/toe marker trajectories provided the stride-level reference MTC. For BMI160 waveform verification, only SPC2019 right-instep IMU signals were used because the custom prototype was mounted on the right shoe, keeping the synchronized MTC-validation branch separate from the unsynchronized BMI160 plausibility branch.

C. Custom BMI160 Walking Recordings

The custom BMI160 recordings were collected using the wearable device described in Section II. Each CSV file contained timestamp, triaxial acceleration, and triaxial angular velocity, matching the main IMU signal types available in SPC2019 [6], [8].

Each recording lasted 30 s in a long, flat corridor. The participant started walking approximately 5 s after recording began, creating a short stationary segment before the walking segment. To match SPC2019, separate slow, normal, and fast

trials were recorded using the participant's preferred walking speed for each condition [6]. Because these recordings were not synchronized with MoCap, they were used only for waveform plausibility rather than direct MTC validation.

D. Trial Conditions and Walking Speeds

For shape verification, the SPC2019 20 m trials were used as the main comparison set because they were closest to the custom corridor recordings [6]. The 10 m trials were kept as additional speed-matched comparisons.

The long trial was excluded from shape verification because it combines all three walking speeds in one continuous recording, making it less suitable for separate slow, normal, and fast waveform comparison [6], [10].

IV. METHODOLOGY

The MTC validation pipeline used the SPC2019 signals defined in Section III and combined a participant-held-out deep-learning gait-event detector, adaptive stride-local inertial processing, virtual toe kinematics, MoCap-referenced toe-offset calibration, and matched IMU-MoCap validation. The gait-event detector is referred to as **Stage A EventNet** in this paper because it represents the first-stage TC/IC detection model used before MTC estimation. Standard IMU clearance steps such as filtering, orientation estimation, integration, and agreement metrics are stated briefly, while the project-specific design choices are described in more detail [2]–[5], [12], [13]. The BMI160 route remained a separate waveform-plausibility check because those recordings were not synchronized with MoCap [10]. The main methodological trade-off was that MoCap-referenced calibration improved offline MTC agreement, but it also prevented the current method from being claimed as a deployment-ready field system.

A. Pipeline Overview and Notation

TC denotes toe-off, IC denotes initial contact, and signed MTC error was defined as IMU estimate minus MoCap reference. MTC values were reported in millimeters. The MoCap toe marker was used as the reference trajectory for stride-level MTC comparison.

B. Dataset Loading and Trial Definition

Each SPC2019 trial was loaded by participant, walking test, and side, with 3.0 s padding added to stabilize filtering and event processing near the analysed region. MoCap gait events were converted to the IMU sample axis so detected IMU events and reference events used the same time base [6], [7].

The evaluated SPC2019 MTC set included slow_10, normal_10, fast_10, slow_20, normal_20, fast_20, and long [6]. The long trial was treated as a continuous mixed-speed walk, so adaptive processing used stride-level gait state rather than speed labels. Results were reported for the 12 completed participants; 54a9 and 8873 were excluded because the strict ML EventNet path produced no valid matched strides before Method 1 toe-offset calibration.

C. Strict-Nested EventNet Gait-Event Detection

The strict-nested EventNet gait-event detection workflow is summarised in Fig. 4.

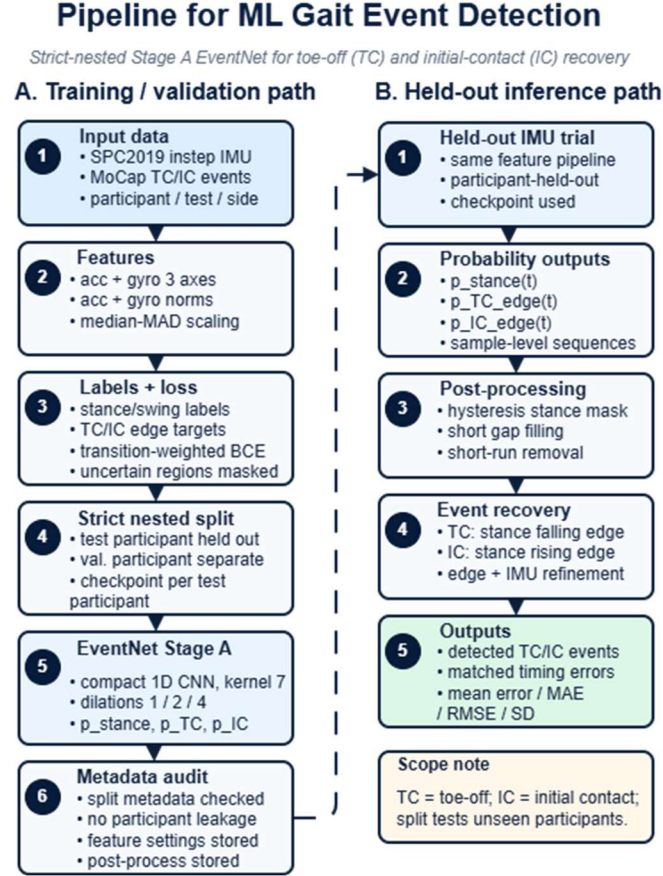


Fig. 4. Strict-nested EventNet gait-event detection pipeline for TC and IC gait-event detection from SPC2019 instep IMU data.

1) Feature Construction

EventNet was trained from instep IMU features: triaxial acceleration, triaxial angular velocity, acceleration norm, and gyroscope norm.

$$\mathbf{x}_t = [\mathbf{a}_t, \boldsymbol{\omega}_t, \|\mathbf{a}_t\|_2, \|\boldsymbol{\omega}_t\|_2] \quad (1)$$

where $\mathbf{a}_t$ is the triaxial acceleration vector, $\boldsymbol{\omega}_t$ is the triaxial angular-velocity vector, and $\mathbf{x}_t$ is the eight-channel EventNet input at sample t . These channels are standard inputs for IMU-based deep-learning gait-event detection [5]. The project specific preparation was the use of per-trial median and median absolute deviation scaling, which made feature normalization more robust to participant- and trial-level amplitude differences before inference.

2) Labels and Transition-Aware Loss

MoCap-derived TC and IC events were converted to the IMU time base and used to generate dense stance/swing labels. Swing was defined over TC-to-IC, reliable stance samples were labelled as stance, and uncertain stride regions were masked from the loss. The nonstandard design choice was the transition-aware EventNet loss: samples near TC and IC were weighted more heavily, and auxiliary TC/IC edge targets were

added so the network learned both gait phase and boundary timing [5].

$$\mathcal{L} = \mathcal{L}_{\text{stance}} + 0.5\mathcal{L}_{\text{TC}} + 0.5\mathcal{L}_{\text{IC}} \quad (2)$$

where $\mathcal{L}_{\text{stance}}$ is the weighted stance/swing binary cross-entropy loss, while $\mathcal{L}_{\text{TC}}$ and $\mathcal{L}_{\text{IC}}$ are auxiliary edge-detection losses around TC and IC. Samples within $\pm 50\text{ms}$ of TC or IC received higher stance-loss weighting, and TC/IC edge targets used $\pm 20\text{ms}$ positive windows.

3) Network Architecture and Strict Nested Splits

The Stage A EventNet model was a compact fully convolutional 1D network with three temporal convolution blocks using kernel size 7 and dilations 1, 2, and 4, followed by a 1x1 output head. This kept the model small while giving enough temporal context for gait-event localization, consistent with convolutional IMU gait-event approaches [5]. In transition-aware mode, the network output stance probability plus TC- and IC-edge probabilities.

Participant leakage was controlled using strict nested splits. For each evaluated participant, a separate checkpoint was trained with that participant excluded from both training and validation; a different participant was used for validation, and all remaining participants formed the training set. The evaluated test/validation pairs were 4d91/5047, 5047/5237, 6e2e/80b8, 9b4b/c9bb, and e54d/4d91. This split design was retained because MTC validation should test generalization to unseen participants, not memorization of participant-specific gait patterns.

4) EventNet Inference and Event Recovery

At inference, the filtered IMU signal was transformed into the same feature set and passed through the participant-held-out checkpoint. The stance-probability sequence was converted into a binary stance mask using hysteresis, short gaps were filled, and unrealistic stance or swing runs were removed. TC and IC were then recovered from stance transitions, with the auxiliary edge heads and local IMU cues used only for boundary refinement [5].

D. Adaptive Stride-Local MTC Estimation

The IMU-based MTC estimation workflow, including stride-local integration, two-stage MoCap-referenced toe-offset calibration, and MoCap validation, is summarised in Fig. 5.

Pipeline for IMU-Based MTC Estimation

Adaptive stride-local vertical integration with Method 1 toe-offset calibration

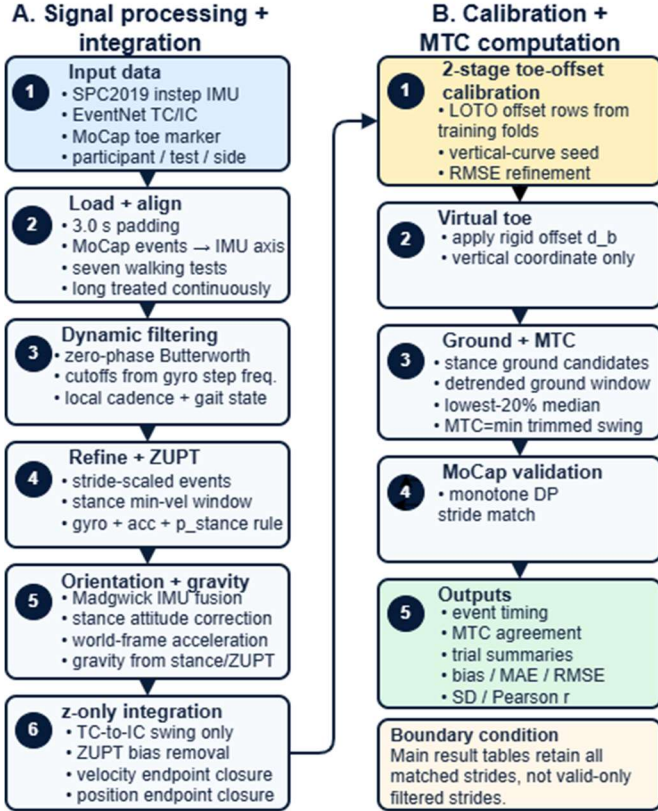


Fig. 5. IMU-based MTC estimation pipeline using adaptive stride-local vertical integration, two-stage MoCap-referenced toe-offset calibration, and MoCap-referenced validation.

1) Dynamic Filtering and Local Gait State

Before event detection and inertial integration, accelerometer and gyroscope channels were filtered with a fourth-order zero-phase Butterworth filter to reduce noise without introducing timing delay [11]. Cadence-adaptive cut-off selection was used: cut-offs were scaled from the gyro-norm step frequency and clipped within bounded ranges, so slow, normal, fast, and long trials were handled using local gait dynamics rather than a single fixed filter setting [6], [7].

$$\begin{aligned} f_{acc} &= \text{clip}(8f_{step}, 10, 35), \\ f_{gyr} &= \text{clip}(6f_{step}, 8, 30) \end{aligned} \quad (3)$$

where f_{step} is the dominant step frequency estimated from the gyroscope-norm spectrum, while f_{acc} and f_{gyr} are the acceleration and gyroscope low-pass cut-off frequencies in Hz.

2) Adaptive Event Refinement and ZUPT Selection

Coarse EventNet TC and IC timings were refined using windows scaled by local stride duration. Slow strides were given wider refinement windows, while fast strides used narrower windows to reduce overcorrection. The local ZUPT or minimum-velocity window was selected inside stance using low gyroscope norm, low acceleration variation, and high EventNet stance probability, following foot-mounted zero-velocity principles but adding stance-probability evidence from the ML detector [12].

3) Orientation, Stance Attitude Correction, and Gravity Removal

Orientation was estimated using the Madgwick IMU filter [13]. Gyroscope values were converted to radians per second, with stronger correction during stance/ZUPT and weaker correction during swing. A conservative stance-attitude correction aligned the estimated gravity direction with measured quiet-stance acceleration, after which acceleration was rotated into the world frame and gravity was removed using stance/ZUPT samples where available.

$$\begin{aligned} a_{w,t} &= R(q_t) a_{b,t}, \\ a_{spec,t} &= a_{w,t} - \hat{g} \end{aligned} \quad (4)$$

where $a_{b,t}$ is body-frame acceleration, $R(q_t)$ is the rotation matrix from the estimated orientation, $a_{w,t}$ is world-frame acceleration, and $\hat{g}$ is the stance/ZUPT gravity estimate.

4) Stride-Local Vertical Integration

The estimator used z-only stride-local integration. For each TC-to-IC swing, vertical acceleration bias was estimated from the local stance/ZUPT window, then vertical velocity and displacement were integrated only across that swing.

$$\begin{aligned} v_{z,t} &= v_{z,t-1} + (a_{spec,z,t} - b_{z,k}) \Delta t, \\ p_{z,t} &= p_{z,t-1} + v_{z,t} \Delta t \end{aligned} \quad (5)$$

where $b_{z,k}$ is the stride-specific vertical acceleration bias estimated from the local stance/ZUPT window for stride k . Endpoint velocity and position were closed at IC to suppress stride-level drift. This design was used because MTC is a vertical clearance quantity and avoiding full-trial 3D trajectory reconstruction reduced exposure to heading drift, horizontal bias, and long-trial accumulation error [2], [12].

5) Virtual Toe Kinematics and MTC Extraction

Toe height was estimated using a calibrated rigid IMU-to-toe offset. The body-frame offset was rotated into the world frame and added to the IMU vertical displacement, which is consistent with foot-mounted IMU clearance methods that estimate heel, toe, or shoe points from sensor-to-foot geometry [3], [4].

$$p_{toe,t} = p_{IMU,t} + R(q_t) d_b, \quad z_{toe,t} = e_z^T p_{toe,t} \quad (6)$$

where d_b is the calibrated IMU-to-toe offset in the sensor body frame, $p_{toe,t}$ is the estimated toe position, and $z_{toe,t}$ is the vertical toe position.

$$MTC_k = \min_{t \in \mathcal{S}_k} (z_{toe,t} - z_{ground,k}) \quad (7)$$

where $\mathcal{S}_k$ is the trimmed swing interval for stride k , and $z_{ground,k}$ is the stride-specific stance-based ground level. Ground height was estimated by robust stance candidates, and the selected window was detrended before using the lowest-sample median to reduce stance-rocker bias before extracting MTC [2], [3].

E. Two-stage MoCap-referenced Toe-offset Calibration

Toe-offset calibration used a two-stage MoCap-referenced calibration procedure. First, a vertical-curve least-squares estimate was used to seed the IMU-to-toe offset. Second, candidate offsets were refined by minimising training-fold IMU-MoCap MTC RMSE, with absolute bias used only as a tiebreaker. Calibration was run separately for each participant,

side, and held-out test, producing leave-one-test-out offset rows that were evaluated using fixed offsets. The search used $x = 0$ to 8 cm, $y = 0$ to 10 cm for the left side or -5 to 5 cm for the right side, $z = -3$ to 3 cm, with no z regularization or forced $z = 0$ constraint. This was needed because MTC was estimated at the toe, not at the IMU origin [3], [4].

The calibration first used a vertical-curve least-squares seed from the IMU-derived swing trajectory. A low-clearance emphasis made the seed more sensitive to the part of swing that controls MTC. Candidate offsets were then refined by scoring training-fold MTC RMSE against the MoCap reference, with absolute bias used only as a tiebreaker [3], [6], [7].

$$d_b^* = \arg \min_{d_b} \sqrt{\frac{1}{N} \sum_{i=1}^N e_i(d_b)^2} \quad (8)$$

where d_b^* is the selected IMU-to-toe offset, e_i is the matched-stride IMU-MoCap MTC error for training-fold stride i , and N is the number of matched training-fold strides. This two-stage design reduced blind grid-search dependence while still choosing the offset that best matched held-out-trial validation.

F. IMU-MoCap Validation

1) Stride Matching and Event Timing

Detected IMU strides were matched to converted MoCap reference strides using monotone dynamic programming based on mid-swing timing. This preserved chronological stride order while allowing missed detections or unmatched references to be skipped [14]. TC and IC timing errors were reported in milliseconds for matched events, and full detected/reference event counts were recovered using the same strict-nested participant-held-out EventNet checkpoints [5].

2) MoCap MTC Reference and Agreement Metrics

MoCap reference MTC was computed from the matched toe-marker trajectory over the swing interval and expressed in millimeters [1], [3], [6], [7]. IMU and MoCap MTC values were joined by matched ground-truth stride ID. Agreement was summarized using standard signed-error, absolute-error, RMSE, SD, and correlation outputs, while the generated artifacts reported event timing, overall MTC agreement, pooled trial summaries, and participant-level diagnostics [2], [3].

G. BMI160 Waveform Verification

The BMI160 waveform-verification workflow is summarised in Fig. 6. This route was kept separate from the MTC validation pipeline because the BMI160 recordings were not synchronized with MoCap.

Pipeline for BMI160 Waveform Verification

Signal-plausibility comparison against SPC2019 right-instep IMU waveforms

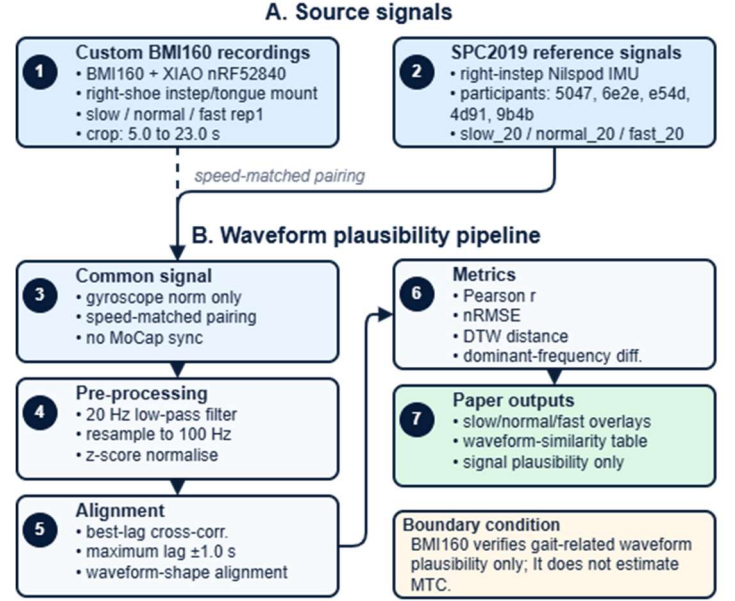


Fig. 6. BMI160 waveform-verification pipeline for signal-plausibility comparison against SPC2019 right-instep IMU waveforms.

1) Signal Preparation

The BMI160 analysis was a waveform plausibility comparison only. It did not perform direct MoCap validation and did not estimate MTC from the BMI160 recordings. The calibrated BMI160 slow, normal, and fast recordings were used. The initial stationary segment was removed by cropping each BMI160 recording from 5.0 s to 23.0 s, and waveform agreement was interpreted only as signal-level similarity rather than biomechanical clearance accuracy [10].

2) Reference Signals and Processing

The BMI160 comparison used SPC2019 right-instep reference signals from the same 12 completed participants used in the final validation cohort [6], [7]. BMI160 slow, normal, and fast rep1 recordings were compared with SPC2019 slow_20, normal_20, and fast_20 respectively using gyroscope norm. The long trial, left-foot signals, and direct BMI160 MTC estimation were excluded from this subsection.

3) Waveform Similarity Metrics

The primary waveform signal was gyroscope norm only. BMI160 and SPC2019 signals were low-pass filtered at 20 Hz where valid, resampled to 100 Hz, z-score normalized, and aligned using best-lag cross-correlation with a maximum lag of ± 1.0 s [11]. Similarity was summarized using Pearson correlation, nRMSE, DTW distance, and dominant-frequency difference; DTW was included because it is a standard dynamic-programming alignment measure for sequences with local timing variation [10], [14].

V. RESULTS

A. ML Gait Event Detection Validation

1) Event Timing Accuracy

TABLE I
Event Timing Accuracy Summary

Event	Reference events, n	Detected events, n	Matched events, n	Detection rate (%)	Mean error (ms)	MAE(ms)
TC	8788	8650	7142	81.27	-9.45	12.46
IC	8788	8650	7142	81.27	-8.61	9.88

Table I summarizes the pooled TC and IC event-detection outputs for the 12 completed participants. Both events had 8788 reference events, 8650 detected events, and 7142 matched events, giving a matched-event detection rate of 81.27%. Mean timing error was -9.448 ms for TC and -8.605 ms for IC, with TC showing larger absolute and spread errors than IC.

2) Event Timing Accuracy by Walking Test

TABLE II
TC Event Timing Accuracy by Walking Test

Metric	slow_10	normal_10	fast_10	slow_20	normal_20	fast_20	long
Reference events, n	672	593	505	694	598	517	5209
Detected events, n	765	646	546	693	614	527	4859
Matched events, n	599	544	447	607	548	460	3937
Detection rate (%)	89.137	91.737	88.515	87.464	91.639	88.975	75.6
Mean error (ms)	-10.491	-10.017	-9.864	-7.827	-7.663	-8.725	-9.75
MAE(ms)	12.089	11.148	10.803	9.597	8.768	9.787	14.1
RMSE(ms)	17.435	16.842	14.808	12.958	11.466	12.874	25.7
SD(ms)	13.937	13.552	11.057	10.336	8.538	9.477	23.8

TABLE III
IC Event Timing Accuracy by Walking Test

Metric	slow_10	normal_10	fast_10	slow_20	normal_20	fast_20	long
Reference events, n	672	593	505	694	598	517	5209
Detected events, n	765	646	546	693	614	527	4859
Matched events, n	599	544	447	607	548	460	3937
Detection rate (%)	89.137	91.737	88.515	87.464	91.639	88.975	75.6
Mean error (ms)	-11.975	-9.658	-8.739	-10.248	-8.982	-8.874	-7.6
MAE(ms)	13.067	10.699	9.875	11.358	9.623	9.596	9.12
RMSE(ms)	17.417	12.676	12.093	13.932	11.39	11.287	13.5
SD(ms)	12.657	8.217	8.369	9.446	7.011	6.983	11.2

Tables II and III report event timing by walking test. The shorter speed-specific trials had matched-event detection rates of approximately 87-92%, whereas the long mixed-speed trial had a lower matched rate of 75.581%. This indicates that unmatched events were mainly driven by long-trial stride matching rather than a uniform event-count failure across all walking tests.

3) Event Timing Error Distribution

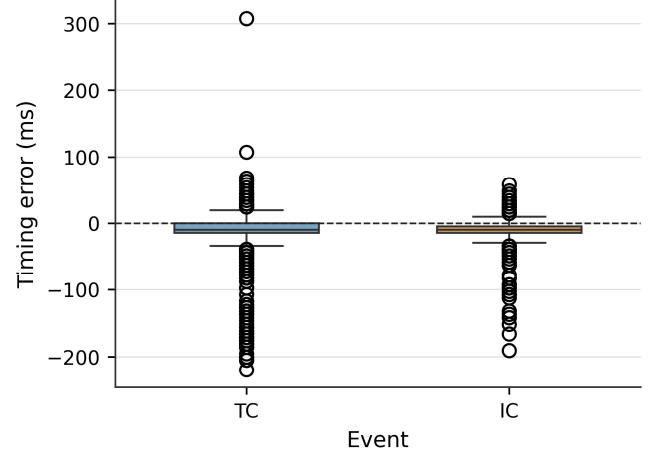


Fig. 7. Distribution of matched TC and IC timing errors from the strict-nested EventNet gait-event detector.

Fig. 7 presents the matched TC and IC timing-error distributions. Both distributions were centered below zero. TC had a wider spread and more negative outliers than IC, while IC errors were more compact.

B. IMU-Based MTC Estimation Validation

1) IMU-MoCap MTC Agreement

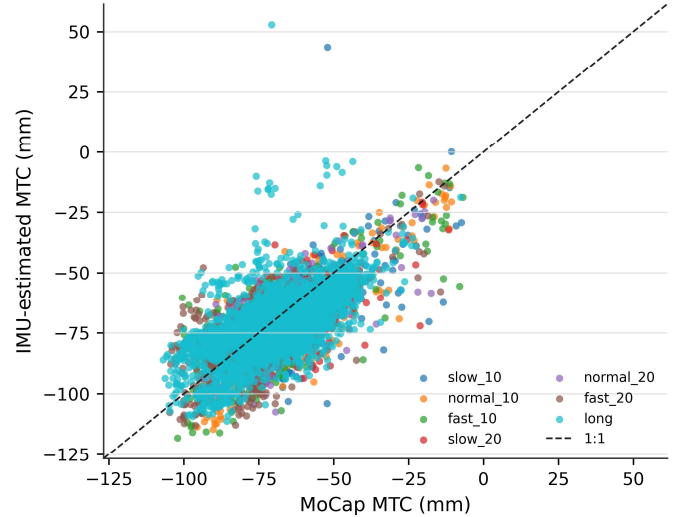


Fig. 8. Agreement between IMU-derived and MoCap-derived MTC values across matched strides.

TABLE IV
Overall IMU-MoCap MTC Agreement Summary

Participants used	Matched strides, n	Mean MoCap MTC (mm)	Mean IMU MTC (mm)	Bias: IMU-MoCap (mm)	MAE (mm)	RMSE (mm)	SD error (mm)	Pearson r
all except for 54a9 and 8873	7142	-67.7	-70.7	-3	8.64	11.4	11	0.67

Fig. 8 and Table IV report the overall IMU-MoCap MTC agreement across all seven walking tests. Across 7142 matched strides, the mean MoCap MTC was -67.696 mm and the mean IMU-estimated MTC was -70.695 mm. The overall bias was -3.000 mm, MAE was 8.638 mm, RMSE was 11.401 mm, and Pearson correlation was $r = 0.674$.

2) Trial-Level Error Metrics

TABLE V
Trial-Level MTC Error Metrics by Walking Test

Trial	slow_10	normal_10	fast_10	slow_20	normal_20	fast_20	long
Strides, n	599	544	447	607	548	460	3937
Mean MoCap MTC (mm)	-60.838	-67.785	-72.157	-61.361	-69.622	-76.228	-67.9
Mean IMU MTC (mm)	-66.185	-70.833	-74.719	-67.076	-71.413	-78.416	-70.5
Bias (mm)	-5.347	-3.048	-2.562	-5.716	-1.791	-2.189	-2.53
MAE (mm)	8.932	8.811	10.578	8.834	7.398	10.42	8.283
RMSE (mm)	12.2	11.423	13.287	11.391	9.653	13.389	11.01
SD error (mm)	10.975	11.019	13.053	9.861	9.494	13.223	10.71
Pearson r	0.631	0.744	0.719	0.556	0.7	0.628	0.645

Table V reports pooled trial-level MTC errors across the 12 completed participants and both sides. The lowest MAE and RMSE occurred in normal_20, at 7.398 mm and 9.653 mm respectively. The highest MAE occurred in fast_10 at 10.578 mm, while the highest RMSE occurred in fast_20 at 13.389 mm. The long mixed-speed trial achieved an RMSE of 11.007 mm.

3) Stride-Level Error Distribution

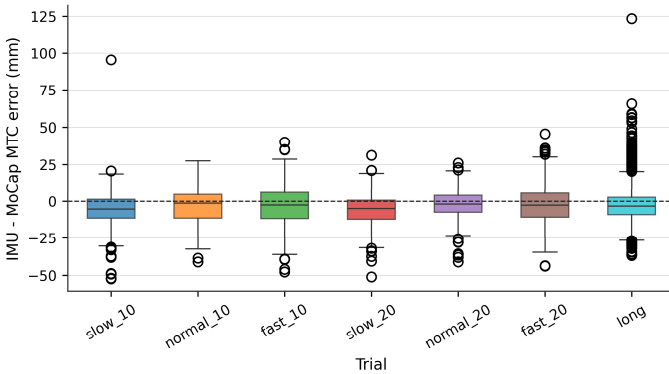


Fig. 9. Boxplot distribution of stride-level IMU-MoCap MTC error across slow, normal, fast, and mixed-speed walking trials.

Fig. 9 presents stride-level MTC error by walking trial. Errors were mostly centered near zero, with wider spread in the fast trials and visible outliers in the long mixed-speed trial.

C. Custom BMI160 Waveform Verification

1) Waveform Shape Comparison

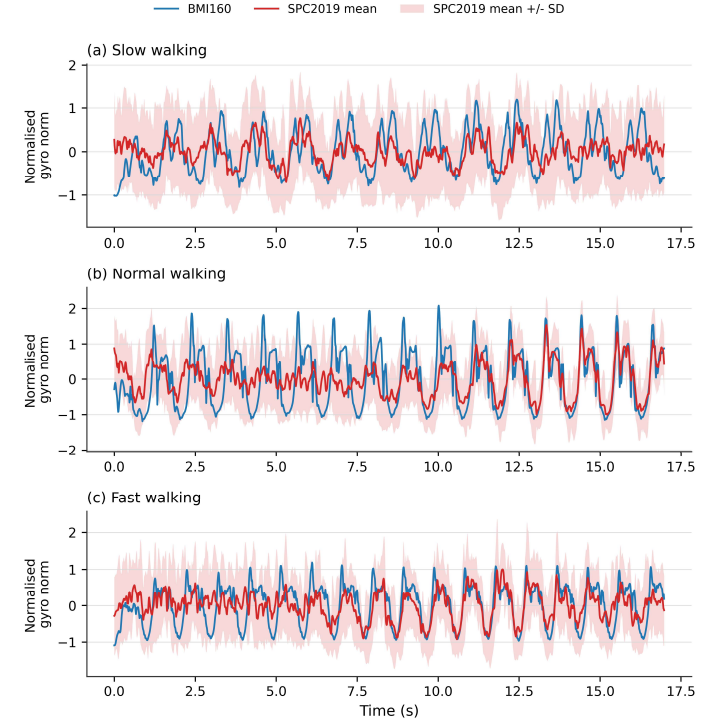


Fig. 10. Multipart waveform overlay comparing custom BMI160 and SPC2019 right-instep reference signals for slow, normal, and fast walking.

Fig. 10 shows repeated gait-cycle peaks in the BMI160 gyroscope-norm waveform for slow, normal, and fast walking. Peak frequency increased from slow to fast walking. The BMI160 waveform did not match every SPC2019 mean peak, but it retained a recognizable periodic walking pattern.

2) Waveform Similarity Metrics

TABLE VI
Waveform Similarity Metrics

BMI160 trial Walking speed	Slow	Normal	Fast
SPC2019 reference trial	slow_20	normal_20	fast_20
Pearson r, mean +/- SD	0.298 +/- 0.243	0.401 +/- 0.176	0.325 +/- 0.154
nRMSE, mean +/- SD	0.309 +/- 0.066	0.294 +/- 0.037	0.284 +/- 0.045
DTW distance, mean +/- SD	0.107 +/- 0.017	0.093 +/- 0.013	0.118 +/- 0.068
Dominant frequency difference (Hz), mean +/- SD	0.052 +/- 0.064	0.021 +/- 0.049	0.062 +/- 0.065

Table VI summarizes waveform similarity between the custom BMI160 recordings and the 12-participant SPC2019 right-instep reference windows. Normal walking showed the strongest similarity, with Pearson $r = 0.401 \pm 0.176$ and $nRMSE = 0.294 \pm 0.037$. Slow walking was the weakest

condition, with Pearson $r = 0.298 \pm 0.243$ and $nRMSE = 0.309 \pm 0.066$. These values support waveform plausibility only and do not validate BMI160-derived MTC.

VI. DISCUSSION

A. MoCap-Referenced MTC Performance

The 12-completed-participant results in Table IV and Fig. 8 show that the strict-nested IMU pipeline estimated MoCap-referenced MTC with millimeter-scale disagreement. The negative bias indicates slight underestimation, while the moderate Pearson correlation suggests that the estimator followed the reference trend but retained participant- and trial-dependent error.

This supports cohort-level feasibility under controlled SPC2019 conditions, but not universal accuracy. Practical use would still require testing across different hardware, footwear, mounting conditions, walking surfaces, and participants [2]-[4], [6], [7].

B. Event-Matching and Trial-Level Behavior

Table I shows that detected event counts were close to reference counts, but fewer events were matched because timing analysis only included TC/IC events paired one-to-one with MoCap strides under the strict monotone gate. Extra, missing, or shifted detections were audited but excluded from timing-error statistics.

Tables II and III show that mismatch was condition-dependent, with shorter speed-specific trials retaining higher matched-event rates than the long mixed-speed trial. Table V further shows that normal_20 gave the most stable MTC agreement, while fast trials produced larger errors, likely due to greater sensitivity to event-boundary placement, ZUPT selection, endpoint correction, and toe-height extraction [2]-[5], [12].

C. Excluded Participants and EventNet Failure Mode

Participants 54a9 and 8873 were excluded because the strict ML EventNet path produced no valid stride matches before Method 1 toe-offset calibration. This was not a post-evaluation RMSE failure; calibration offsets could not be fitted. Diagnostics showed that IMU-only detection recovered many affected strides, while the selected ML checkpoints produced sparse or phase-shifted event sequences that failed the fixed MoCap matching gate.

This suggests that validation BCE loss alone may select checkpoints with acceptable stance-mask loss but poor TC/IC recovery. Future strict-nested training should therefore use validation TC/IC timing metrics and matched-event F1 for checkpoint selection.

D. Bias and Error Mechanisms

The negative MTC bias is likely cumulative rather than caused by one isolated block. Slightly early TC/IC detection can shift the TC-to-IC swing segment used to locate minimum toe height. Residual orientation error and gravity-removal uncertainty may also accumulate during vertical double

integration, despite Madgwick filtering, stance correction, ZUPT/minimum-velocity information, and endpoint closure [2]-[4], [12], [13].

The calibrated virtual toe offset adds further uncertainty. Although two-stage MoCap-referenced calibration reduced participant- and side-specific geometry mismatch, the instep-IMU-to-toe relationship was still modelled as a rigid offset. Shoe deformation, strap tension, casing movement, soft-tissue artefact, and marker-to-sensor coordinate differences may therefore shift the estimated toe-height curve [3], [4], [7].

E. BMI160 Waveform Plausibility

The BMI160 branch should be interpreted only as waveform plausibility evidence. The repeated gyroscope-norm peaks in Fig. 10 show that the BMI160/XIAO prototype captured periodic walking-related inertial signals, while the frequency increase from slow to fast walking supports basic speed responsiveness [8], [9].

However, Table VI does not validate BMI160-derived MTC. Pearson correlation, $nRMSE$, lag-aligned comparison, and DTW assess waveform shape, not toe-clearance accuracy [10], [14]. The device should therefore be described as signal-plausible, not MTC-validated, until synchronized BMI160-MoCap testing is completed.

F. Limitations and Future Work

The main limitation is that each validation branch was constrained by a different evidence level. The SPC2019 branch allowed synchronized IMU-MoCap MTC validation, but only for participants and trials where strict-nested EventNet produced valid matched strides. The BMI160 branch tested the intended low-cost wearable hardware but lacked synchronized MoCap reference data and therefore could support only signal plausibility, not MTC accuracy.

The second limitation is calibration dependency. The two-stage MoCap-referenced toe-offset calibration is suitable for offline dataset validation but not field deployment. A practical system should replace this with direct calibration, anthropometric estimation, shoe-geometry modelling, or another calibration-free approach.

The third limitation is the BMI160 validation level. The custom device was tested with one participant and without synchronized optical reference, so the current evidence supports raw-signal plausibility only. The required next experiment is a synchronized BMI160-MoCap protocol with fixed mounting and toe-marker reference.

Future work should focus on event-timing checkpoint selection, fast-walking and long-trial robustness, improved ZUPT selection, stance-attitude correction, vertical endpoint constraints, and calibration-free toe-offset estimation. Hardware work should also quantify BLE reliability, onboard buffering, CSV logging, battery life, enclosure stability, and repeated shoe-mounting consistency before applied deployment is claimed.

VII. CONCLUSION

This study evaluated two separate parts of a low-cost wearable MTC workflow: an IMU-based MTC algorithm validated using SPC2019 MoCap data, and a BMI160/XIAO prototype verified at waveform level. The strict-nested algorithm was validated on 12 SPC2019 participants, producing 7142 matched strides with a bias of -3.000 mm, MAE of 8.638 mm, RMSE of 11.401 mm, and Pearson correlation of $r = 0.674$. This shows millimeter-scale agreement under controlled conditions, particularly during normal-speed walking.

However, accuracy remained dependent on event timing, drift correction, ZUPT selection, gravity removal, sensor-to-toe geometry, and MoCap-referenced toe-offset calibration. Fast and long mixed-speed trials were more error-sensitive, and two participants were excluded because strict ML event recovery failed before calibration.

The BMI160/XIAO prototype captured repeatable gyroscope-norm gait waveforms and speed-dependent rhythm changes, supporting signal plausibility only. Because the BMI160 recordings were not synchronized with MoCap, the prototype is not yet validated for MTC accuracy. Overall, the work should be positioned as a validated offline algorithm with a signal-plausible low-cost prototype, with synchronized BMI160-MoCap testing required as the next step.

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